

Lab 06

Name:

江承恩

Student ID:

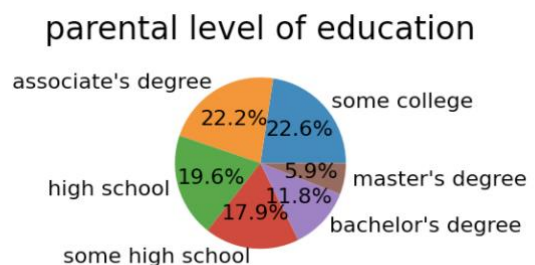
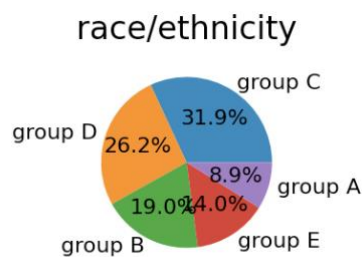
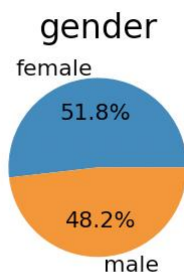
B09611004

Total Score:

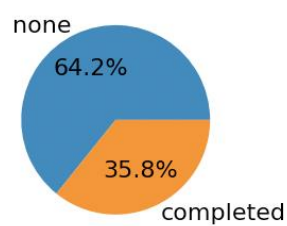
1. (a) It contains : "gender", "race/ethnicity", "parental level of education", "lunch", "test preparation course", "math score", "reading score", "writing score".

(b) There isn't any missing value. However, if there are some missing values in the data, we probably can consider not using that piece of data, or, filling in with an appropriate value.

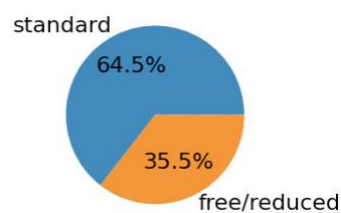
(c) In this case, the number of outliers aren't large enough, they can be ignored. Quoted on the Internet: "the "outlier" is actually the only datum "putting things right". Removing it would unnecessarily introduce bias", which can check my thoughts.
2. (a)
 - **Gender:** female is a little more than male but the difference is small.
 - **Race/ethnicity:** group C and group D accounted for half and group A is a relatively small group.
 - **Parental level of education:** comprehensively, bachelor and master are in the minority.
 - **Test preparation course:** none completed is about two times larger than completed.
 - **Lunch:** standard is also about two times larger than free/reduced.



test preparation course

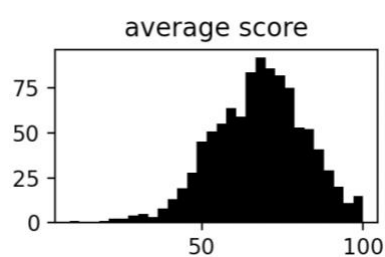
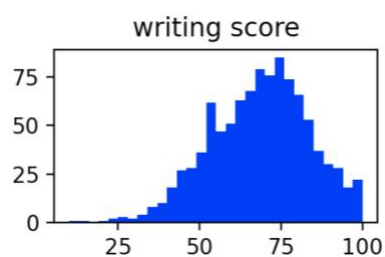
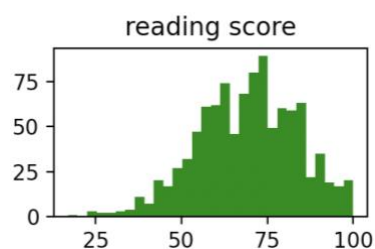
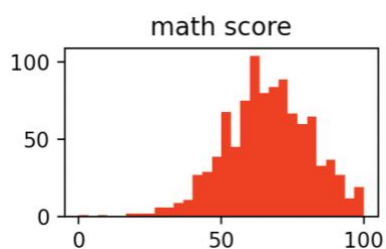


lunch



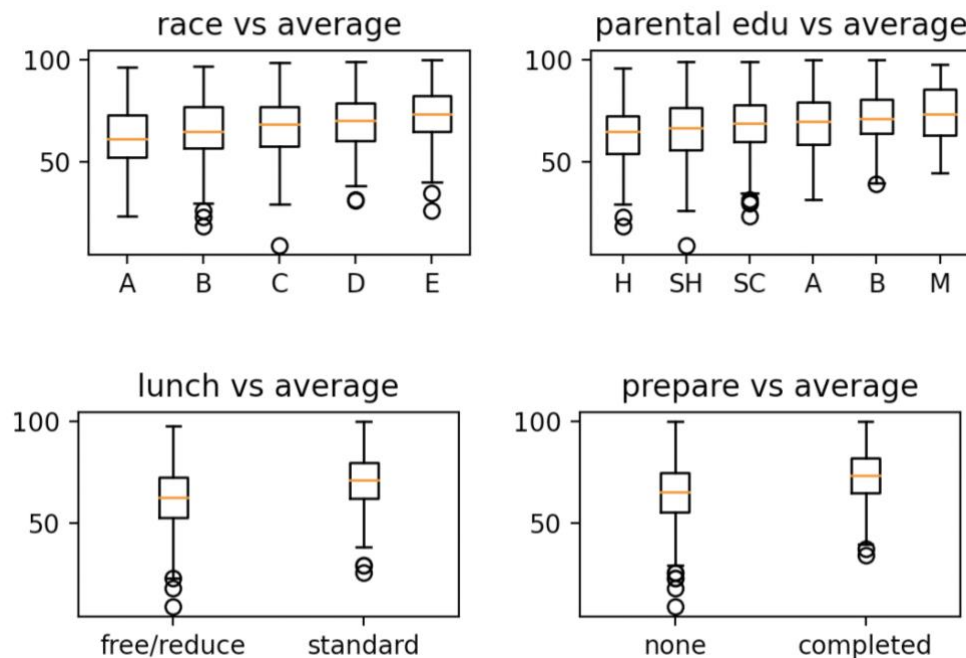
(b)

	MATH	READING	WRITING	AVERAGE
MEAN	66.09	69.17	68.05	67.77
STD	15.16	14.6	15.19	14.25
MIN	0	17	10	9
25%	57	59	57.75	58.33
50%	66	70	69	68.33
75%	77	79	79	77.66
MAX	100	100	100	100



3. (a)

- **Race v.s. average:** the average score is increasing by group changed alphabetically
- **Parental education v.s. average:** although master and bachelor are the minority, but they earn the highest average score. So, the average score has positive correlation with parental education.
- **Lunch v.s. average:** we can see that average score has positive correlation with amount of lunch.
- **Test preparation v.s. average:** we can also see that average score has positive correlation with the test preparation completed or not.



(b)

- **Gender v.s. lunch v.s. average:** We can see female both got higher average score in standard or free/reduced than male.
- **Gender v.s. test preparation v.s. average:** we also can see female both got higher average score in none or completed than male.

